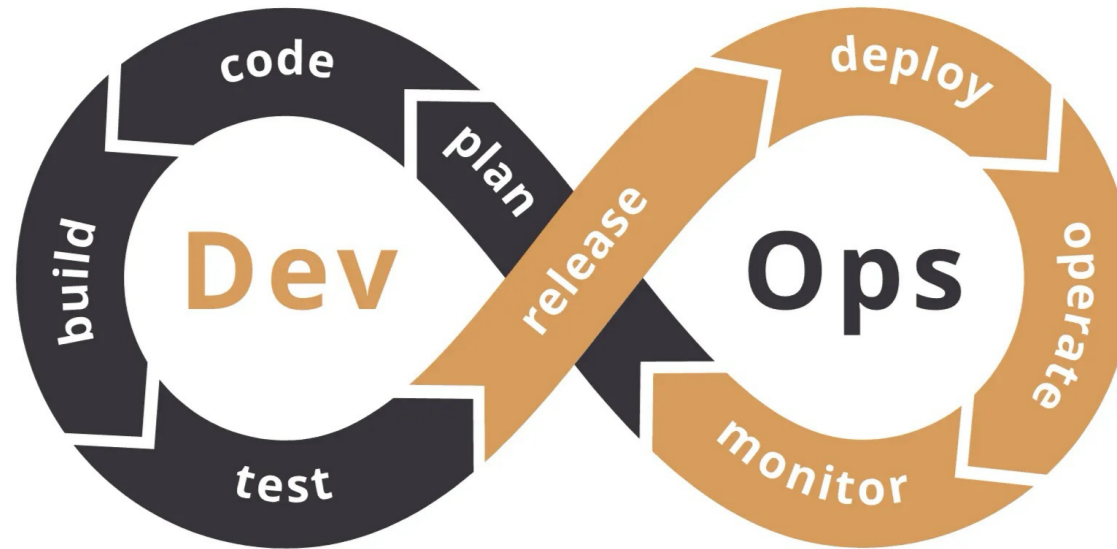


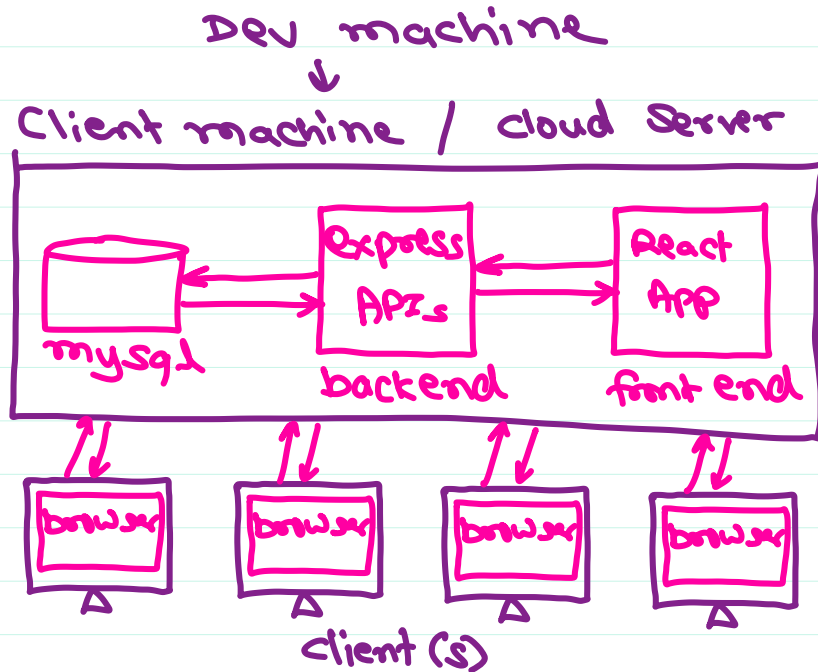


Trainer: Nilesh Ghule

Application Deployment

$appn = code + runtime + libs + conf + os$

① multiple apps run on single machine



✓ cost effective

✗ SPOF

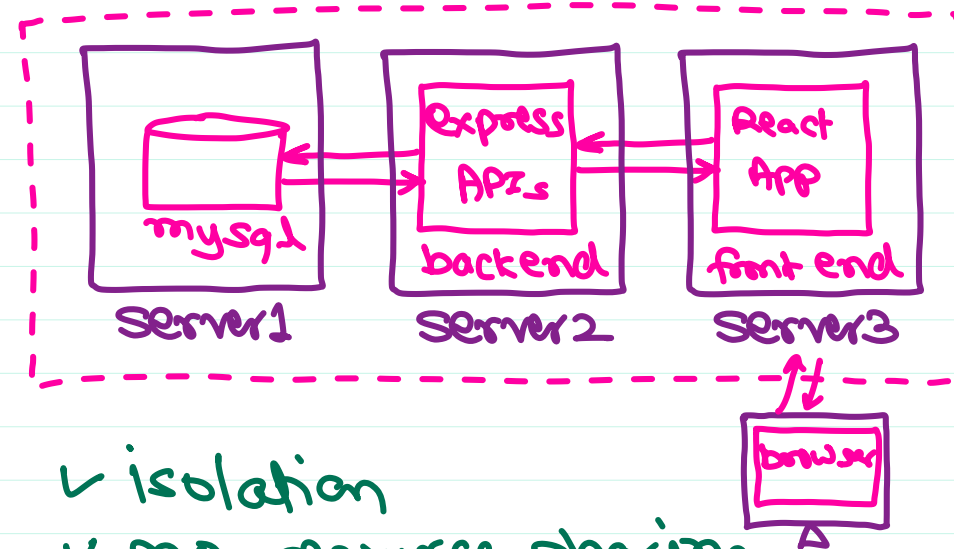
✗ resource sharing

✗ works on my machine

② multiple apps run on multiple machines

* Capital expenditure

* operational expenditure



✓ isolation

✓ no resource sharing

✗ Cost increases

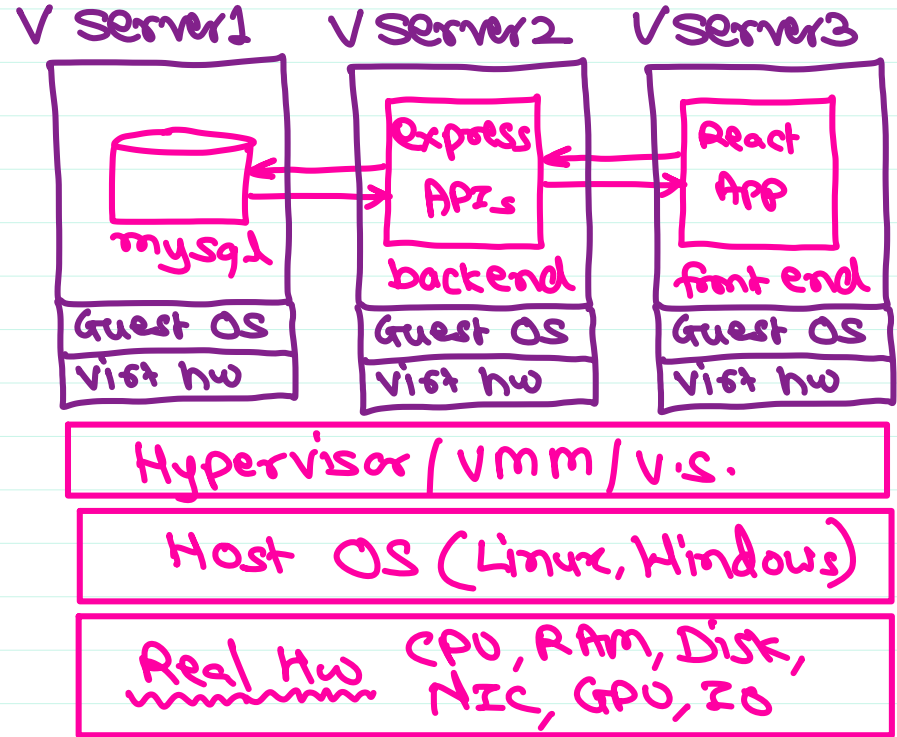
- separate machines ↑

- operational cost ↑

Virtualisation → vCPU, vGPU, vMemory, vIO, vDisk, vNIC, ...

③ hardware virtualization - Type II - Hypervisor

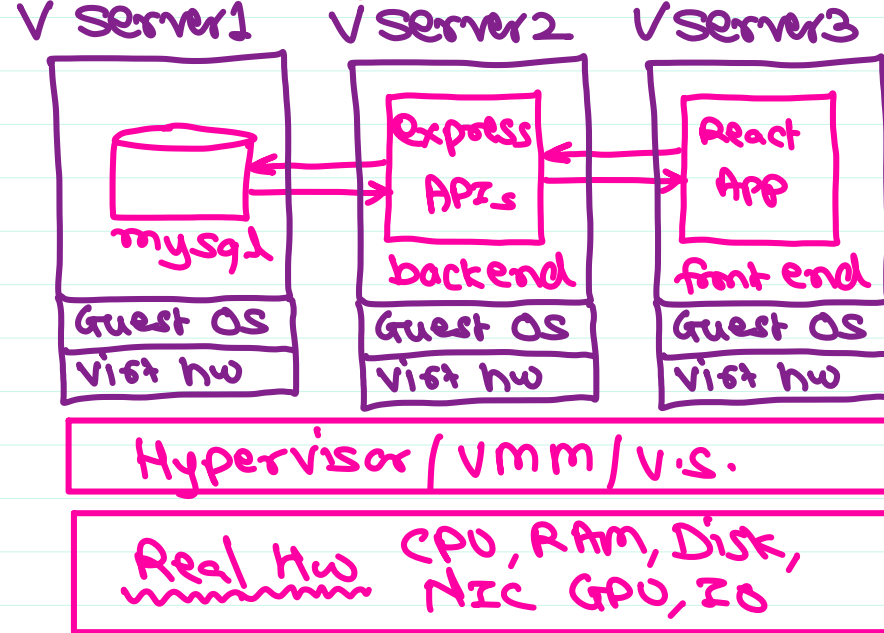
* Capital expenditure ↓



* advantages

- lower cost
- isolation
- backup
- deployment

④ hardware virtualization - Type I - Hypervisor



Vmware ESXi

* Data centers

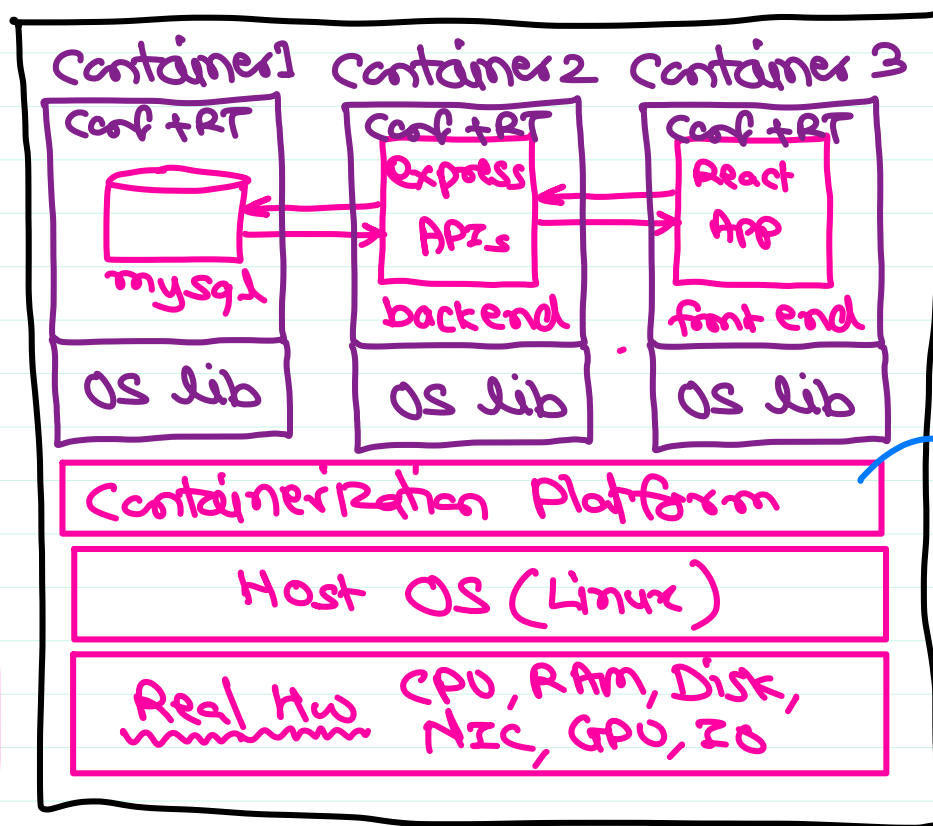
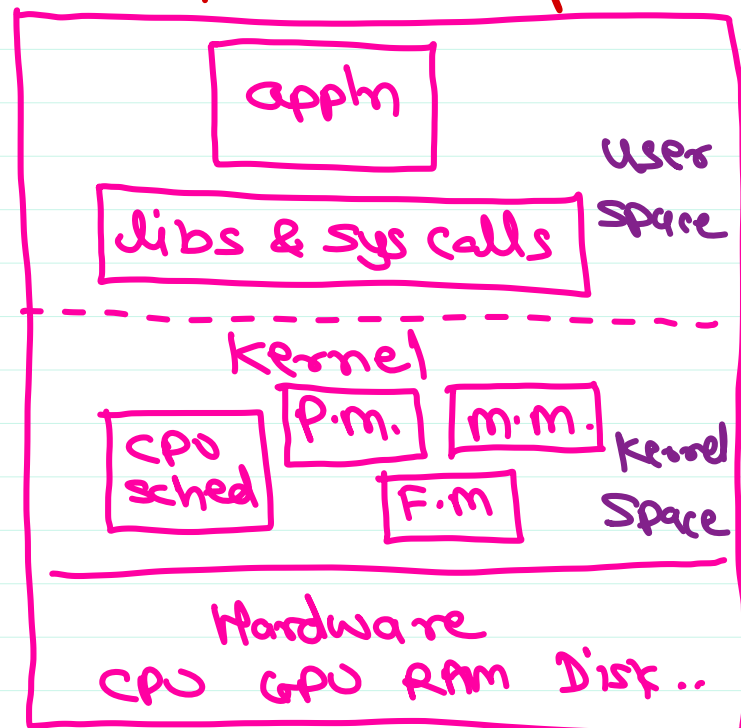
* limitations

- heavy OS
- boot time
- resource usage (mem, disk, cpu)
- OS duplication
- OS mgmt & licensing

Containerisation - OS virtualization

⑤ OS user space vs kernel space

⑥ Containerization



vm	Container
Full OS	Shared Kernel
Heavy	Light weight
Slow boot	Fast startup
GBs	MBs
Hw virt.	OS isolation

Lxc
docker
podman
CRI-O

→ achieved with Linux features:

- ① namespaces
- ② cgroups

* advantages

- lower cost

- isolation

- backup
- deployment
- OS sharing
- no extra OS licensing & maintenance cost.

appm = code + dependencies + config + runtime + OS dep.

Container → appm exec environment

Docker

- Open-source containerization platform.
- Enables
 - App deployment
 - App isolation
 - App scaling
 - Security
 - Efficient exec.

History

- ① Solomon Hykes
2008 - dotCloud - PaaS
- Use Linux Containers to solve "works on my machine".
- ② Complex LXC to Simplified Docker (2013).
Docker = Dock Worker who handle Cargo Containers.
Open Sourced to Apache.

* Docker Installation 2
official docs.



* Docker in action

> docker run hello-world

> docker run -d -p 8080:80 httpd

* Image

✓ read-only multi-layer template

✓ Immutable

✓ From DockerFile

✓ Shared via reg.

→ Container

✓ an instance of image

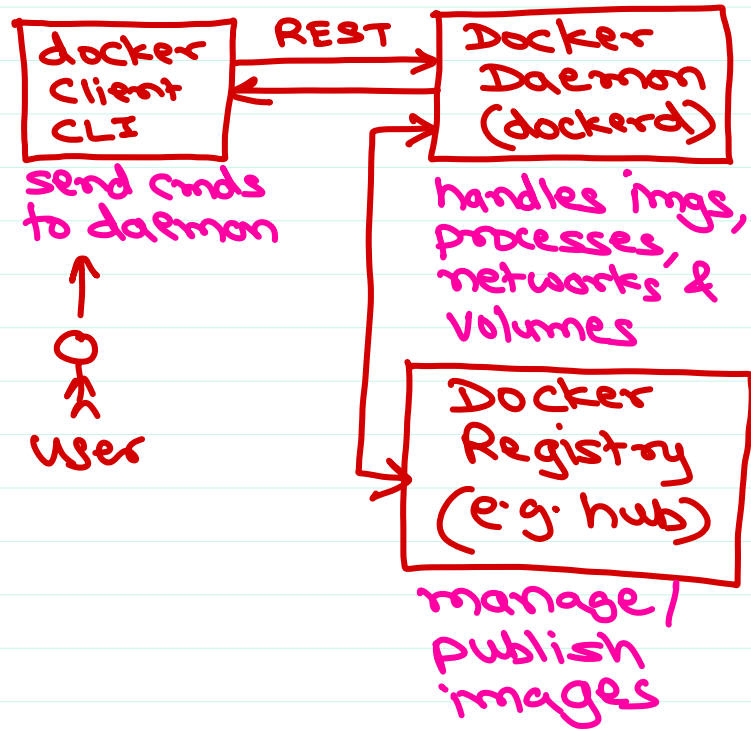
✓ Thin writable layer

✓ From Image

✓ Start, Stop, Destroy

Docker

* Docker Architecture



Container Lifecycle

my-apache ← index.html

* PS [-a]

① create

cp index.html

② start

③ pause

④ unpause

⑤ stop

logs

⑥ rm [-f]

docker images

* images

① pull

② ~~run~~ = create + start

③ rmi

④ build from Docker file

⑤ push (?)



Thank You!

Nilesh Ghule

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